

* Rank 0 tensor.

$$[M]$$

$$M = M_s^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & M_y & 0 \end{bmatrix}$$

$$M_y = M_{py}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & M_{xz} \\ 0 & M_{yy} & 0 \\ M_{xz} & 0 & M_{zz} \end{bmatrix}$$

$$M_{xx} = -M_{du}^{(1)} + M_{dv}^{(1)} + M_s^{(1)}$$

$$M_{xz} = M_{dxz}^{(1)}$$

$$M_{yy} = -M_{du}^{(1)} - M_{dv}^{(1)} + M_s^{(1)}$$

$$M_{zz} = 2M_{du}^{(1)} + M_s^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ -M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = -T_{py}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & M_{yyy} & 0 \\ 0 & M_{zzz} & 0 \\ M_{yzx} & 0 & M_{zzz} \\ 0 & M_{yzx} & 0 \\ M_{xxy} & 0 & M_{yzx} \end{bmatrix}$$

$$M_{xxy} = -M_{fay}^{(1)} - M_{fby}^{(1)} + M_{py}^{(1)}$$

$$M_{yyy} = 2M_{fay}^{(1)} + 3M_{py}^{(1)}$$

$$M_{zzz} = -M_{fay}^{(1)} + M_{fby}^{(1)} + M_{py}^{(1)}$$

$$M_{yzx} = M_{f3}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzz} & 0 \\ M_{yzx} & 0 & -\frac{M_{zzz}}{2} \\ 0 & M_{xxy} & 0 \\ -\frac{M_{xxy}}{2} & 0 & -M_{yzx} - M_{xxy} \end{bmatrix}$$

$$M_{xxy} = 2M_{py}^{(2)} - 2T_{dxz}^{(1)}$$

$$M_{zzy} = 2M_{py}^{(2)} + 2T_{dxz}^{(1)}$$

$$M_{yzx} = 3T_{du}^{(1)} + T_{dv}^{(1)}$$

$$\frac{M_{zzy}}{2} = M_{py}^{(2)} + T_{dxz}^{(1)}$$

$$M_{zxy} = -3T_{du}^{(1)} + T_{dv}^{(1)}$$

$$\frac{M_{xxy}}{2} = M_{py}^{(2)} - T_{dxz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & M_{yzz} \\ 0 & M_{zxy} & 0 \\ M_{xyx} & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = -T_{du}^{(2)} + T_{dv}^{(2)} + T_s^{(1)}$$

$$M_{yzz} = -M_{py}^{(3)} + T_{dxz}^{(2)}$$

$$M_{zxy} = -T_{du}^{(2)} - T_{dv}^{(2)} + T_s^{(1)}$$

$$M_{xyx} = M_{py}^{(3)} + T_{dxz}^{(2)}$$

$$M_{xyz} = 2T_{du}^{(2)} + T_s^{(1)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & M_{xxzx} & 0 \\ M_{xxyy} & M_{yyyy} & M_{yyzz} & 0 & M_{yyzx} & 0 \\ M_{xxzz} & M_{yyzz} & M_{zzzz} & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} \\ M_{xxzx} & M_{yyzx} & M_{zzzx} & 0 & M_{zxzx} & 0 \\ 0 & 0 & 0 & M_{yzxy} & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = -2M_{du}^{(1)} - 4M_{dv}^{(2)} + 2M_{du}^{(1)} + 4M_{dv}^{(2)} - M_{gu}^{(1)} + M_{gv}^{(1)} + 2M_g^{(1)} + M_s^{(1)} + 2M_s^{(2)}$$

$$M_{xxyy} = -2M_{du}^{(1)} + 2M_{gu}^{(1)} - M_g^{(1)} + M_s^{(1)}$$

$$M_{xxzz} = M_{du}^{(1)} + M_{dv}^{(1)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)}$$

$$M_{xxzx} = M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{gay}^{(1)} - M_{gby}^{(1)}$$

$$M_{yyyy} = -2M_{du}^{(1)} - 4M_{dv}^{(2)} - 2M_{dv}^{(1)} - 4M_{dv}^{(2)} - M_{gu}^{(1)} - M_{gv}^{(1)} + 2M_g^{(1)} + M_s^{(1)} + 2M_s^{(2)}$$

$$M_{yyzz} = M_{du}^{(1)} - M_{dv}^{(1)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)}$$

$$M_{yyzx} = M_{dxz}^{(1)} + 2M_{gby}^{(1)}$$

$$M_{zzzz} = 4M_{du}^{(1)} + 8M_{du}^{(2)} + 2M_{gu}^{(1)} + 2M_g^{(1)} + M_s^{(1)} + 2M_s^{(2)}$$

$$M_{zzzx} = M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{gay}^{(1)} - M_{gby}^{(1)}$$

$$M_{yzyz} = M_{du}^{(2)} - M_{dv}^{(2)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(2)}$$

$$M_{yzxy} = M_{dxz}^{(2)} + 2M_{gby}^{(1)}$$

$$M_{zxzx} = M_{du}^{(2)} + M_{dv}^{(2)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(2)}$$

$$M_{xyxy} = -2M_{du}^{(2)} + 2M_{gu}^{(1)} - M_g^{(1)} + M_s^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & M_{xxzx} & 0 \\ -M_{xxyy} & 0 & M_{yyzz} & 0 & M_{yyzx} & 0 \\ -M_{xxzz} & -M_{yyzz} & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ -M_{xxzx} & -M_{yyzx} & -M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxyy} &= -2M_{dv}^{(3)} + 4T_{f3}^{(1)} \\
M_{xxzz} &= 3M_{du}^{(3)} - M_{dv}^{(3)} - 4T_{f3}^{(1)} \\
M_{xxzx} &= M_{dxz}^{(3)} + 2T_{fay}^{(1)} + 2T_{fby}^{(1)} - 2T_{py}^{(1)} \\
M_{yyzz} &= 3M_{du}^{(3)} + M_{dv}^{(3)} + 4T_{f3}^{(1)} \\
M_{yyzx} &= M_{dxz}^{(3)} - 4T_{fby}^{(1)} \\
M_{zzzx} &= M_{dxz}^{(3)} - 2T_{fay}^{(1)} + 2T_{fby}^{(1)} + 2T_{py}^{(1)} \\
M_{yzxy} &= 4T_{fay}^{(1)} + T_{py}^{(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & M_{yzxy} \\ 0 & M_{zxzx} & 0 \\ M_{yzxy} & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= -2M_{du}^{(4)} + 2M_{dv}^{(4)} + M_s^{(3)} \\
M_{yzxy} &= 2M_{dxz}^{(4)} \\
M_{zxzx} &= -2M_{du}^{(4)} - 2M_{dv}^{(4)} + M_s^{(3)} \\
M_{xyxy} &= 4M_{du}^{(4)} + M_s^{(3)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = T_{py}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & 0 \\ 0 & M_{yyzx} & 0 \\ 0 & M_{zzzx} & 0 \\ M_{yzyz} & 0 & M_{yzxy} \\ 0 & M_{zxzx} & 0 \\ M_{xyyz} & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{xxzx} &= 2M_{dxz}^{(5)} - T_{fay}^{(2)} - T_{fby}^{(2)} + T_{py}^{(3)} \\
M_{yyzx} &= 2T_{fay}^{(2)} + T_{py}^{(3)} + 2T_{py}^{(4)} \\
M_{zzzx} &= -2M_{dxz}^{(5)} - T_{fay}^{(2)} + T_{fby}^{(2)} + T_{py}^{(3)} \\
M_{yzyz} &= -3M_{du}^{(5)} - M_{dv}^{(5)} + T_{f3}^{(2)} \\
M_{yzxy} &= M_{dxz}^{(5)} - T_{fay}^{(2)} + T_{fby}^{(2)} + T_{py}^{(4)} \\
M_{zxzx} &= 3M_{du}^{(5)} - M_{dv}^{(5)} + T_{f3}^{(2)} \\
M_{xyyz} &= -M_{dxz}^{(5)} - T_{fay}^{(2)} - T_{fby}^{(2)} + T_{py}^{(4)} \\
M_{xyxy} &= 2M_{dv}^{(5)} + T_{f3}^{(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} \\ M_{zxzx} & M_{zxyy} & M_{zxzz} & 0 & M_{zxzx} & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= -3M_{du}^{(6)} - M_{dv}^{(6)} + T_{f3}^{(3)} \\
M_{yzxy} &= -M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(6)} \\
M_{zxxx} &= 2M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{zxyy} &= 2T_{fay}^{(3)} + T_{py}^{(5)} + 2T_{py}^{(6)} \\
M_{zxxz} &= -2M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{zxxz} &= 3M_{du}^{(6)} - M_{dv}^{(6)} + T_{f3}^{(3)} \\
M_{xyyz} &= M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(6)} \\
M_{xyxy} &= 2M_{dv}^{(6)} + T_{f3}^{(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & M_{xxxx} & 0 & 0 & M_{xxxx} & 0 \\
M_{yyxx} & M_{yyyy} & M_{yyzz} & 0 & M_{yyzx} & 0 & 0 & M_{yyxz} & 0 \\
M_{zzxx} & M_{zzyy} & M_{zzzz} & 0 & M_{zzzx} & 0 & 0 & M_{zzxz} & 0 \\
0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} & M_{yzzy} & 0 & M_{yzyx} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & M_{zxxz} & 0 & 0 & M_{zxxz} & 0 \\
0 & 0 & 0 & M_{xyyz} & 0 & M_{xyxy} & M_{xyzy} & 0 & M_{xyyx}
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= -2M_{du}^{(1)} - 4M_{du}^{(2)} + 2M_{dv}^{(1)} + 4M_{dv}^{(2)} - M_{gu}^{(1)} + M_{gv}^{(1)} + 2M_g^{(1)} + M_s^{(1)} + 2M_s^{(2)} \\
M_{xxyy} &= -2M_{du}^{(1)} - 2M_{dv}^{(3)} + 2M_{gu}^{(1)} - M_g^{(1)} + M_s^{(1)} + 4T_{f3}^{(1)} \\
M_{xxzz} &= M_{du}^{(1)} + 3M_{du}^{(3)} + M_{dv}^{(1)} - M_{dv}^{(3)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} - 4T_{f3}^{(1)} \\
M_{xxzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} + 2M_{dxz}^{(5)} - M_{gay}^{(1)} - M_{gby}^{(1)} + 2T_{fay}^{(1)} - T_{fay}^{(2)} + 2T_{fby}^{(1)} - T_{fby}^{(2)} - 2T_{py}^{(1)} + T_{py}^{(3)} \\
M_{xxxz} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} - M_{gay}^{(1)} - M_{gby}^{(1)} + 2T_{fay}^{(1)} + T_{fay}^{(2)} + 2T_{fby}^{(1)} + T_{fby}^{(2)} - 2T_{py}^{(1)} - T_{py}^{(3)} \\
M_{yyxx} &= -2M_{du}^{(1)} + 2M_{dv}^{(3)} + 2M_{gu}^{(1)} - M_g^{(1)} + M_s^{(1)} - 4T_{f3}^{(1)} \\
M_{yyyy} &= -2M_{du}^{(1)} - 4M_{du}^{(2)} - 2M_{dv}^{(1)} - 4M_{dv}^{(2)} - M_{gu}^{(1)} - M_{gv}^{(1)} + 2M_g^{(1)} + M_s^{(1)} + 2M_s^{(2)} \\
M_{yyzz} &= M_{du}^{(1)} + 3M_{du}^{(3)} - M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} + 4T_{f3}^{(1)} \\
M_{yyzx} &= M_{dxz}^{(1)} + M_{dxz}^{(3)} + 2M_{gby}^{(1)} + 2T_{fay}^{(2)} - 4T_{fby}^{(1)} + T_{py}^{(3)} + 2T_{py}^{(4)} \\
M_{yyxz} &= M_{dxz}^{(1)} + M_{dxz}^{(3)} + 2M_{gby}^{(1)} - 2T_{fay}^{(2)} - 4T_{fby}^{(1)} - T_{py}^{(3)} - 2T_{py}^{(4)} \\
M_{zzxx} &= M_{du}^{(1)} - 3M_{du}^{(3)} + M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} + 4T_{f3}^{(1)} \\
M_{zzyy} &= M_{du}^{(1)} - 3M_{du}^{(3)} - M_{dv}^{(1)} - M_{dv}^{(3)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} - 4T_{f3}^{(1)} \\
M_{zzzz} &= 4M_{du}^{(1)} + 8M_{du}^{(2)} + 2M_{gu}^{(1)} + 2M_g^{(1)} + M_s^{(1)} + 2M_s^{(2)} \\
M_{zzzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} - 2T_{fay}^{(1)} - T_{fay}^{(2)} + 2T_{fby}^{(1)} + T_{fby}^{(2)} + 2T_{py}^{(1)} + T_{py}^{(3)} \\
M_{zzxz} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} + 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} - 2T_{fay}^{(1)} + T_{fay}^{(2)} + 2T_{fby}^{(1)} - T_{fby}^{(2)} + 2T_{py}^{(1)} - T_{py}^{(3)} \\
M_{yzyz} &= M_{du}^{(2)} - 3M_{du}^{(5)} - M_{dv}^{(2)} - M_{dv}^{(5)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(2)} + T_{f3}^{(2)} \\
M_{yzxy} &= M_{dxz}^{(2)} + M_{dxz}^{(5)} + 2M_{gby}^{(1)} + 4T_{fay}^{(1)} - T_{fay}^{(2)} + T_{fby}^{(2)} + T_{py}^{(1)} + T_{py}^{(4)} \\
M_{yzzy} &= M_{du}^{(2)} + 3M_{du}^{(5)} - M_{dv}^{(2)} + M_{dv}^{(5)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(2)} - T_{f3}^{(2)} \\
M_{zyzx} &= M_{dxz}^{(2)} - M_{dxz}^{(5)} + 2M_{gby}^{(1)} + 4T_{fay}^{(1)} + T_{fay}^{(2)} - T_{fby}^{(2)} + T_{py}^{(1)} - T_{py}^{(4)} \\
M_{zxxx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{dxz}^{(3)} - M_{gay}^{(1)} - M_{gby}^{(1)} - 2T_{fay}^{(1)} - 2T_{fby}^{(1)} + 2T_{py}^{(1)} \\
M_{zxyy} &= M_{dxz}^{(1)} - M_{dxz}^{(3)} + 2M_{gby}^{(1)} + 4T_{fby}^{(1)} \\
M_{zxxx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{dxz}^{(3)} + M_{gay}^{(1)} - M_{gby}^{(1)} + 2T_{fay}^{(1)} - 2T_{fby}^{(1)} - 2T_{py}^{(1)} \\
M_{zxzx} &= M_{du}^{(2)} + 3M_{du}^{(5)} + M_{dv}^{(2)} - M_{dv}^{(5)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(2)} + T_{f3}^{(2)} \\
M_{zxxz} &= M_{du}^{(2)} - 3M_{du}^{(5)} + M_{dv}^{(2)} + M_{dv}^{(5)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(2)} - T_{f3}^{(2)} \\
M_{xyyz} &= M_{dxz}^{(2)} - M_{dxz}^{(5)} + 2M_{gby}^{(1)} - 4T_{fay}^{(1)} - T_{fay}^{(2)} - T_{fby}^{(2)} - T_{py}^{(1)} + T_{py}^{(4)} \\
M_{xyxy} &= -2M_{du}^{(2)} + 2M_{dv}^{(5)} + 2M_{gu}^{(1)} - M_g^{(1)} + M_s^{(2)} + T_{f3}^{(2)} \\
M_{xyzy} &= M_{dxz}^{(2)} + M_{dxz}^{(5)} + 2M_{gby}^{(1)} - 4T_{fay}^{(1)} + T_{fay}^{(2)} + T_{fby}^{(2)} - T_{py}^{(1)} - T_{py}^{(4)} \\
M_{xyyx} &= -2M_{du}^{(2)} - 2M_{dv}^{(5)} + 2M_{gu}^{(1)} - M_g^{(1)} + M_s^{(2)} - T_{f3}^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} & M_{yzzy} & 0 & M_{yzyx} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & M_{zxzx} & 0 & 0 & M_{zxzx} & 0 \\
0 & 0 & 0 & M_{xyyz} & 0 & M_{xyxy} & M_{xyzy} & 0 & M_{xyyx}
\end{bmatrix}$$

$$\begin{aligned}
M_{yzyz} &= -2M_{du}^{(4)} - 3M_{du}^{(6)} + 2M_{dv}^{(4)} - M_{dv}^{(6)} + M_s^{(3)} + T_{f3}^{(3)} \\
M_{yzxy} &= 2M_{dxz}^{(4)} - M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(2)} + T_{py}^{(6)} \\
M_{yzyy} &= 2M_{du}^{(4)} - 3M_{du}^{(6)} - 2M_{dv}^{(4)} - M_{dv}^{(6)} - M_s^{(3)} + T_{f3}^{(3)} \\
M_{yzyx} &= -2M_{dxz}^{(4)} - M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} - T_{py}^{(2)} + T_{py}^{(6)} \\
M_{zxxx} &= 2M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{zxyy} &= 2T_{fay}^{(3)} + T_{py}^{(5)} + 2T_{py}^{(6)} \\
M_{zxxz} &= -2M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{zxxz} &= -2M_{du}^{(4)} + 3M_{du}^{(6)} - 2M_{dv}^{(4)} - M_{dv}^{(6)} + M_s^{(3)} + T_{f3}^{(3)} \\
M_{zxxz} &= 2M_{du}^{(4)} + 3M_{du}^{(6)} + 2M_{dv}^{(4)} - M_{dv}^{(6)} - M_s^{(3)} + T_{f3}^{(3)} \\
M_{xyyz} &= 2M_{dxz}^{(4)} + M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} - T_{py}^{(2)} + T_{py}^{(6)} \\
M_{xyxy} &= 4M_{du}^{(4)} + 2M_{dv}^{(6)} + M_s^{(3)} + T_{f3}^{(3)} \\
M_{xyzy} &= -2M_{dxz}^{(4)} + M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(2)} + T_{py}^{(6)} \\
M_{xyyx} &= -4M_{du}^{(4)} + 2M_{dv}^{(6)} - M_s^{(3)} + T_{f3}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
M_{xxxx} & 0 & M_{xxxx} \\
0 & M_{yyyy} & 0 \\
M_{zzzx} & 0 & M_{zzzx} \\
M_{yyyz} & 0 & M_{yyyz} \\
M_{zzxx} & 0 & M_{zzxx} \\
0 & M_{xxyy} & 0 \\
0 & M_{yzyy} & 0 \\
M_{xxxx} & 0 & M_{xxxx} \\
M_{xxyy} + M_{yyzz} - M_{yzzy} - M_{zxxz} + M_{zzxx} & 0 & M_{xxyy} \\
0 & M_{xyyz} + \frac{M_{zzxz}}{2} - \frac{M_{zzzx}}{2} & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= -2M_{du}^{(1)} - 4M_{du}^{(2)} + 2M_{dv}^{(1)} + 4M_{dv}^{(2)} - M_{gu}^{(1)} + M_{gv}^{(1)} + 2M_g^{(1)} + 3M_s^{(1)} \\
M_{xxzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} - M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{yyyy} &= -2M_{du}^{(1)} - 4M_{du}^{(2)} - 2M_{dv}^{(1)} - 4M_{dv}^{(2)} - M_{gu}^{(1)} - M_{gv}^{(1)} + 2M_g^{(1)} + 3M_s^{(1)} \\
M_{zzzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{zzzz} &= 4M_{du}^{(1)} + 8M_{du}^{(2)} + 2M_{gu}^{(1)} + 2M_g^{(1)} + 3M_s^{(1)} \\
M_{yyyz} &= M_{dxz}^{(1)} + M_{dxz}^{(3)} + 2M_{gby}^{(1)} \\
M_{yyyz} &= M_{du}^{(1)} + 3M_{du}^{(3)} - M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} \\
M_{zzxx} &= M_{du}^{(1)} - 3M_{du}^{(3)} + M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} \\
M_{zzzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} + 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{xxyy} &= -2M_{du}^{(1)} - 2M_{dv}^{(3)} + 2M_{gu}^{(1)} - M_g^{(1)} + M_s^{(1)} \\
M_{yzzy} &= M_{du}^{(2)} + 3M_{du}^{(5)} - M_{dv}^{(2)} + M_{dv}^{(5)} - M_{gu}^{(1)} + M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} \\
M_{zxxx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{dxz}^{(3)} - M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{zxxz} &= M_{du}^{(2)} - 3M_{du}^{(5)} + M_{dv}^{(2)} + M_{dv}^{(5)} - M_{gu}^{(1)} - M_{gv}^{(1)} - M_g^{(1)} + M_s^{(1)} \\
M_{xyyz} &= M_{dxz}^{(2)} - M_{dxz}^{(5)} + 2M_{gby}^{(1)}
\end{aligned}$$